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Abbreviations and Acronyms

Air tugger	(air-operated hoist)
ALARP	As Low As Reasonably Practicable
ASAP	As Soon as Possible
BHA	Bottom Hole Assembly
BOPE	Blowout Preventer Equipment
Chevron	(Chevron U.S.A., Inc.)
Chevron's Handbook	(Chevron's GOM Business Units Shelf and Deepwater Contractor's Handbook – Revised January 2008)
CIBP	Cast Iron Bridge Plug
CPR	Cardio Pulmonary Resuscitation
Driller	Hercules Offshore Rig 120 Driller
FH-1	Hercules Offshore Rig 120 Floor Hand No. 1 (the Deceased)
FH-2	Hercules Offshore Rig 120 Floor Hand No. 2
FH-3	Hercules Offshore Rig 120 Floor Hand No. 3
FIT	Formation Integrity Test
Flow Chart	(Events/Causal Factor Diagram of the Accident)
GOM	Gulf of Mexico
GR-CCL	Gamma Ray-Casing Collar Locator
HCR	Remotely Operated Hydraulic Control Valve
HO	Hercules Offshore Rig 120 Hoist Operator
HSE	Health, Environmental and Safety
IV	(Intravenous fluids)
JHA	Job Hazard Analysis
JSA	Job Safety Analysis
MES	Multi-Linear Event Sequence Diagram
MMS	Minerals Management Service
MOL	Move on Location
MP	Main Pass
ND	Nipple-Down
NU	Nipple-Up

OCS	Outer Continental Shelf
OIM	Hercules Offshore Installation Manager/Licensed Master (Captain) in command of a liftboat
OSHA	Occupational Safety and Health Administration
OSM	(MMS Regional Office of Safety Management)
PPE	Personal Protective Equipment
ppg	pounds per gallon
PIC	Person-in-Charge
POOH	Pull Out of Hole
SBM	Synthetic Base Mud
SLS	Secondary Load Support
Subpart O	(MMS 30 CFR Part 250, Subpart O training requirements)
SWA	Stop Work Authority
SWL	Maximum Safe Working Load
T&A'd	Temporarily Abandoned
The rig	(Hercules Offshore Rig 120)
The well	(Main Pass Block 30, Lease OCS-G 4903 Well No. A-14 ST01)
TIH	Trip in Hole
USCG	United States Coast Guard
VETCO	(Established in July 2004 through subsidiaries Vetco Gray and Abiel. Vetco Gray was sold to General Electric on January 9, 2007)

Executive Summary

An accident that resulted in one fatality and an injury occurred on the Hercules Offshore Rig 120 (the rig) contracted by Chevron USA Inc. (Chevron) to conduct sidetracking operations from Lease OCS-G-4903, Main Pass (MP) Block 30, Well No. A-14 ST01 (the well), in the Gulf of Mexico (GOM) offshore Louisiana, on 7 March 2009 at approximately 2158 hours. Lifting operations from the rig floor were in progress utilizing the rig floor air tugger (air-operated hoist) to position a 2- inch rubber hose, with internal steel construction rated for high pressure, for testing the well's 13-3/8 inch x 9-7/8 inch casing annulus. The fifty (50) feet, approximately 400 pounds, of hose assembly was lowered from the rig floor to the wellhead deck where the lower end of the hose assembly would be connected into the well's 13-3/8 inch x 9-7/8 inch casing annulus valve and the upper end of the hose assembly connected to the rig floor manifold. Subsequent to lowering the hose assembly from the rig floor to near the wellhead deck, it was determined that the 50 feet of hose assembly failed to reach the well's annular casing valve by approximately 10 feet. At this time the Hercules Offshore Hoist Operator (Hoist Operator) began to retrieve the hose assembly back into the derrick above the rig floor where it could be swapped-out with 100 feet of hose section that was located on one wall of the rig floor. The lift connection being used was comprised of a WECO Figure 1502 hammer union that had been previously modified by the addition of a welded bale. When the 50 feet of hose assembly was lifted approximately 30 feet into the derrick above the rig floor, the lift connection failed just above the weld on one end of the bale. The Hercules Offshore Floor Hand No. 1 (FH-1), standing near the center of the rig floor but not directly under the assembly, received life threatening injuries (unresponsive but breathing) subsequent to being struck on the top of his hardhat by the hose assembly. The hose assembly then continued its descent from the rig floor's edge through the rig's air gap and next to the wellhead deck into the GOM. The Hercules Offshore Rig Floor Hand No. 2 (FH-2), standing next to the hand railing on the rig floor and manipulating the hose to protect the load from snagging on any protrusions, was initially knocked down by the hose. FH-2 arose to move away from the rig floor railing, only to be knocked down again by the air-operated hoist wire rope that back-lashed subsequent to the loss of the load's weight. FH-2 sustained a superficial laceration to his right triceps with right shoulder pain.

The Minerals Management Service (MMS) accident investigation Panel concluded that the accident was preventable. The **Cause** of the 7 March 2009 accident was the use of a modified WECO Figure 1502 hammer union that failed just above the weld on one end of the welded bale. **Possible Contributing**

Causes include: (1) improper monitoring of the Hercules Offshore equipment inspection program, (2) lift crew location during the bulk hose lifting operation, (3) the lack of a formal written Job Safety Analysis (JSA) to accompany the pre-job lift safety meeting and (4) JSA policy oversight by rig management.

Introduction

Authority

A fatal accident and injury occurred on 7 *March 2009* at approximately *2158 hours* aboard the jack-up drilling rig Hercules Offshore Rig 120 (the rig) contracted to Chevron USA Inc. (Chevron) while operations were being conducted on Lease OCS-G 4903, Main Pass (MP) Block 30, Well No. A-14 ST01 (the well), in the Gulf of Mexico (GOM) offshore Louisiana.

The fatally injured person was a Hercules Offshore Floor Hand (FH-1) employee. At the time of the accident FH-1 was observing a bulk hose lifting operation from the near the center of the rig floor but not directly under the hose assembly. FH-1 had just changed positions with another Hercules Offshore Floor Hand (FH-2) who was standing next to the rig floor hand rail and manipulating the hose to protect the load from snagging on any protrusions.

Pursuant to 43 U.S.C. 1348(d)(1) and (2) and (f) [Outer Continental Shelf (OCS) Lands Act, as amended] and Department of the Interior regulations 30 CFR 250, the Minerals Management Service (MMS) is required to investigate and prepare a report of this accident. By memorandum dated 13 March 2009, the following personnel were named to the investigative panel:

Glynn T. Breaux, Chairman – Office of Safety Management, Field Operations, GOM OCS Region
Randy Josey – Office of Safety Management, Field Operations, GOM OCS Region
David Emilien – New Orleans District, Field Operations, GOM OCS Region

Background

Lease OCS-G 4903 covers approximately 5,000 acres and is located in Main Pass Block 30, Gulf of Mexico, off the Louisiana Coast (*see Figure 1*). The lease was purchased by seven (7) original Lessees in Sale Number 66, with an effective lease date of 1 December 1981 and a first producing lease date of 28 November 1987. Chevron U.S.A. Inc. is the record title interest owner effective 8 August 2005 and became designated operator on 21 June 2006.